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COMPUTATIONAL STATISTICS EXAM (ASTA 607)

QUESTION ONE(a)

The crabs data frame consists of the Morphological Measurements on Leptograpsus Crabs with 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species Leptograpsus variegatus. The data frame was collected at Fremantle, W. Australia for a multivariate study of variation in two species of rock crab of genus Leptograpsus by Campbell, N.A and Mahon, R.J in 1974.

QUESTION ONE(b)

#Loading the dataset

Code:

```
crabs
head(crabs)
```

Output:

```
sp sex index   FL  RW   CL   CW  BD
1  B   M      1  8.1 6.7 16.1 19.0 7.0
2  B   M      2  8.8 7.7 18.1 20.8 7.4
3  B   M      3  9.2 7.8 19.0 22.4 7.7
4  B   M      4  9.6 7.9 20.1 23.1 8.2
5  B   M      5  9.8 8.0 20.3 23.0 8.2
6  B   M      6 10.8 9.0 23.0 26.5 9.8
```

#Structure of the dataset

Code:

```
str(crabs)
```

Output:

```
'data.frame':  200 obs. of  8 variables:
 $ sp   : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 ...
 $ sex  : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 ...
```

```
 $ index: int  1 2 3 4 5 6 7 8 9 10 ...
 $ FL   : num  8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...
 $ RW   : num  6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
 $ CL   : num  16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2
...
 $ CW   : num  19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3
...
 $ BD   : num  7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
>
```

Interpretation:

The crabs data is a data frame with 200 observations and 8 variables.

The variables and their types in the data frame are:

- Species - factor
- Sex - factor
- Index - integer
- Frontal Lobe size (mm) - numeric
- Rear Width (mm) - numeric
- Carapace Width (mm) - numeric
- Body Depth (mm) – numeric

QUESTION ONE(c)

#Frequency table for species

Code:

```
crab_species<-factor(x=crabs$sp, labels = c("Blue", "Orange"))
species_table<-table(crab_species)
```

Output:

```
crab_species
  Blue Orange
   100    100
```

Interpretation:

The frequency table displays 100 blue crab species and 100 orange crab species

#Frequency table for sex

Code:

```
crab_sex<-factor(x=crabs$sex, labels = c("Female", "Male"))
sex_table<-table(crab_sex)
```

Output:

```
Female   Male
   100    100
```

Interpretation:

The frequency table displays 100 female crabs and 100 male crabs.

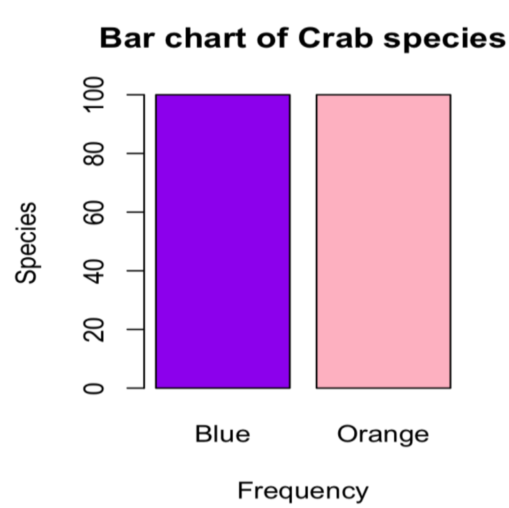
QUESTION ONE(d)

#Code:

```
barplot(species_table,main = "Bar chart of Crab species", ylab
= "Species",
        xlab = "Frequency", col=c("purple","pink"))
```

#Barplot for Group composition - Species

Output:



Interpretation:

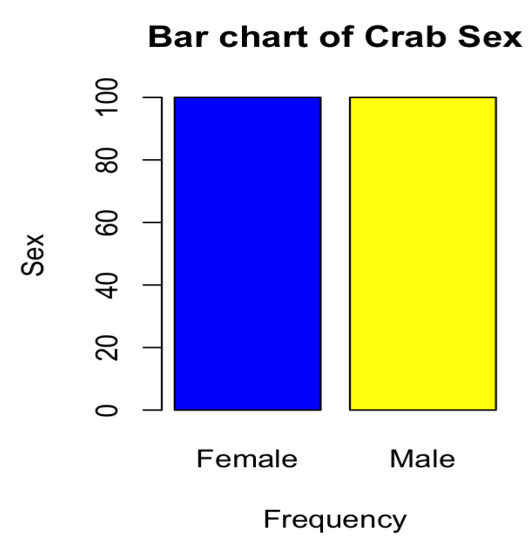
The number of blue and orange species are the same.

#Barplot for Group composition - Sex

Code:

```
barplot(sex_table,main = "Bar chart of Crab Sex", ylab = "Sex",
        xlab = "Frequency", col=c("blue","yellow"))
```

Output:



Interpretation:

The number of female and male species are the same.

QUESTION ONE(e)

The dataset is balanced amongst the four groups because the species are divided into equal parts as well as the sex this helps with statistical analysis.

QUESTION TWO (a)

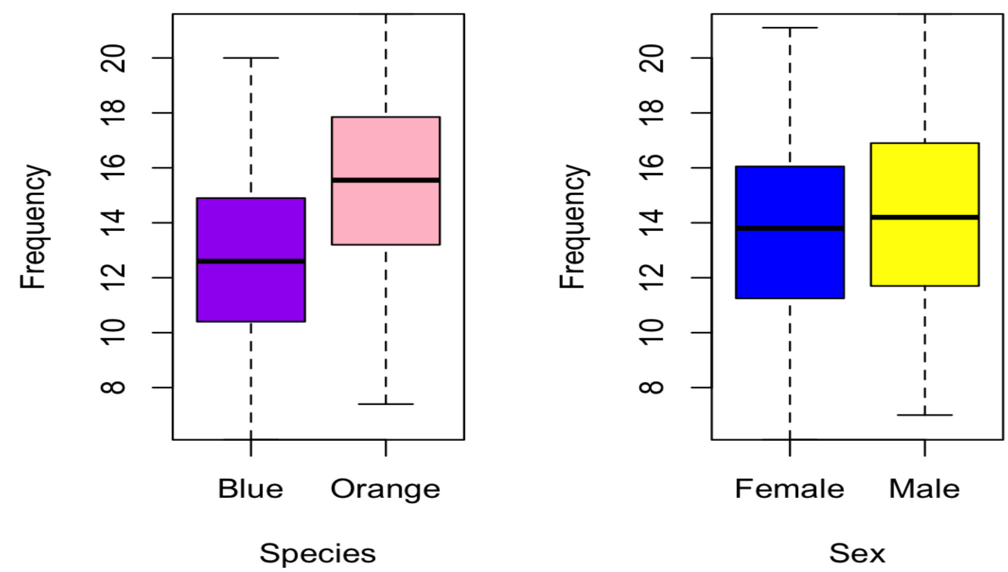
#Side-By-Side boxplot of body depth for the four groups by species and sex

Code:

```
crabs$sp<-crab_species
```

```
crabs$sex<-crab_sex
par(mfrow = c(1,2))
boxplot(BD~sp, data = crabs,col=c("purple","pink") )
boxplot(BD~sex, data = crabs,col=c("blue","yellow") )
```

Output:



QUESTION TWO (b)

<u>Species</u>

There are zero to no outliers. The blue and orange crabs are both widely spread. The median is also in the center which implies that the distribution of the species are approximately normal.

<u>Sex</u>

There are zero to no outliers. The female and male crabs are both widely spread. The median is also in the center which implies that the distribution of the sex is approximately normal.

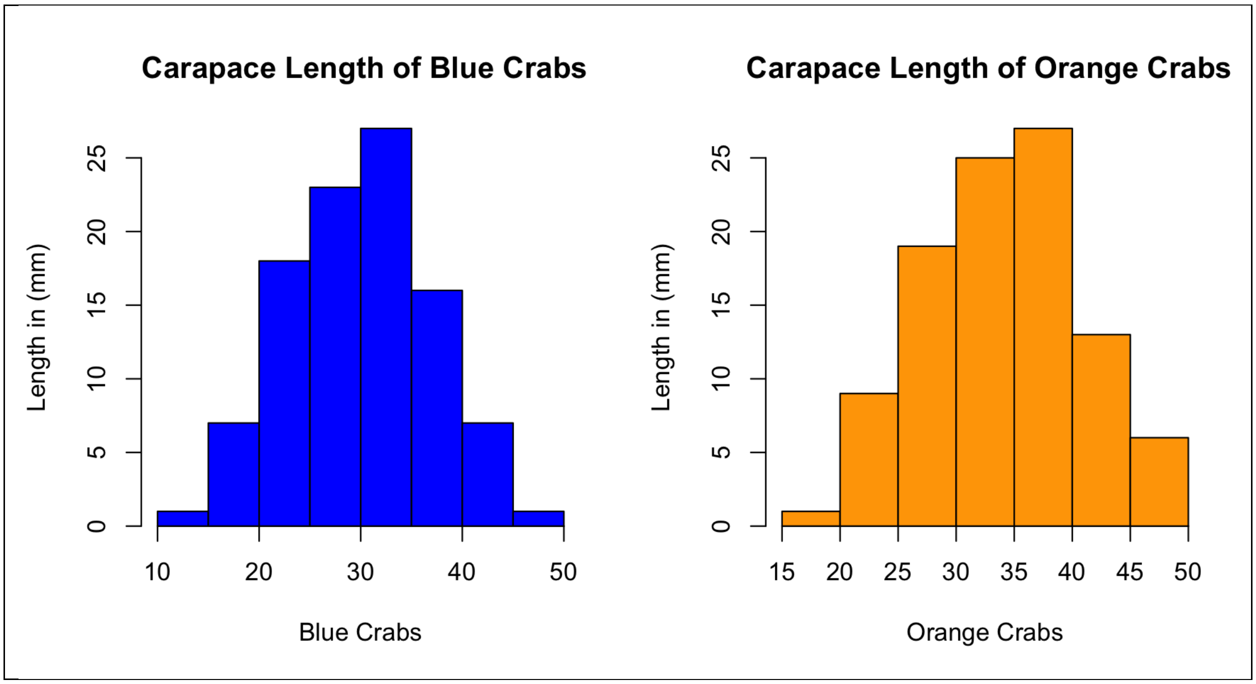
QUESTION THREE (a)

#Histogram for each species

Code:

```
par(mfrow = c(1,2))
hist(x=crabs$CL[crabs == "Blue"], xlab = "Blue Crabs",
     ylab = "Length in (mm)", col = "blue",
     main = " Carapace Length of Blue Crabs")

hist(x=crabs$CL[crabs == "Orange"],xlab = "Orange Crabs",
     ylab = "Length in (mm)",col = "orange",
     main = " Carapace Length of Orange Crabs")
```

Output:



QUESTION THREE (b)

Blue Crabs

The carapace length for blue crabs are approximately normally distributed with a single mode of.

Orange Crabs

The carapace length for orange crabs is a bit skewed to the left with a single mode.

QUESTION FOUR (a)

#Mean of each morphological measurement by species

Code:

```
mean_by_species<-crabs %>%
  group_by(sp) %>%
  summarise(
    mean_FL = mean(FL),
    mean_RW = mean(RW),
    mean_CL = mean(CL),
    mean_CW = mean(CW),
    mean_BD = mean(BD)
  )
```

Output:

```
sp      mean_FL mean_RW mean_CL mean_CW mean_BD
  <fct>    <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
1 Blue      14.1    11.9    30.1    34.7    12.6
2 Orange    17.1    13.5    34.2    38.1    15.5
>
```

#Mean of each morphological measurement by sex

Code:

```
mean_by_sex<-crabs %>%
  group_by(sex) %>%
  summarise(
    mean_FL = mean(FL),
    mean_RW = mean(RW),
    mean_CL = mean(CL),
    mean_CW = mean(CW),
    mean_BD = mean(BD)
  )
```

Output:

```
sex     mean_FL mean_RW mean_CL mean_CW mean_BD
  <fct>    <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
1 Female    15.4    13.5    31.4    35.8    13.7
2 Male      15.7    12.0    32.9    37.0    14.3
```

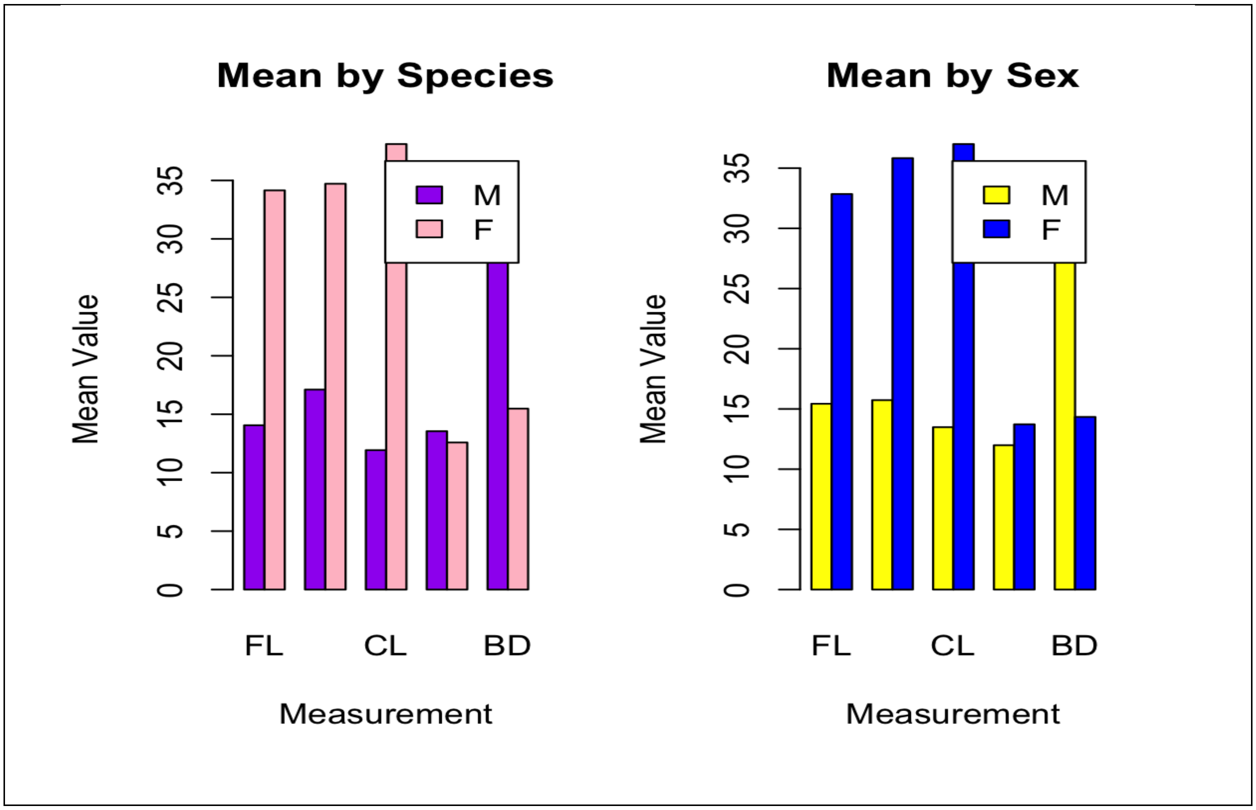
QUESTION FOUR (b)

#Barplot of mean of morphological measurements by Sex and Species

Code:

```
#Barplot of mean of morphological measurements by Sex

mean_by_sex_matrix <- matrix(
  c(mean_by_sex$mean_FL, mean_by_sex$mean_RW,
mean_by_sex$mean_CL,
    mean_by_sex$mean_CW, mean_by_sex$mean_BD),
  nrow = 2, byrow = TRUE,
  dimnames = list(sex = c("M", "F"),
                  measurement = c("FL", "RW", "CL", "CW",
"BD"))
)

barplot(mean_by_sex_matrix, beside = TRUE,
        col = c("yellow", "blue"),
        legend = TRUE,
        main = "Mean by Sex",
        xlab = "Measurement",
        ylab = "Mean Value")

mean_by_species_matrix <- matrix(
  c(mean_by_species$mean_FL, mean_by_species$mean_RW,
mean_by_species$mean_CL,
    mean_by_species$mean_CW, mean_by_species$mean_BD),
  nrow = 2, byrow = TRUE,
  dimnames = list(sex = c("M", "F"),
                  measurement = c("FL", "RW", "CL", "CW",
"BD"))
)

#Barplot of mean of morphological measurements by Species
barplot(mean_by_species_matrix, beside = TRUE,
        col = c("purple", "pink"),
        legend = TRUE,
        main = "Mean by Sex",
        xlab = "Measurement",
        ylab = "Mean Value")
```

Output:



QUESTION FOUR (c)

Great species variation : The Carapace Length show the great species variation.

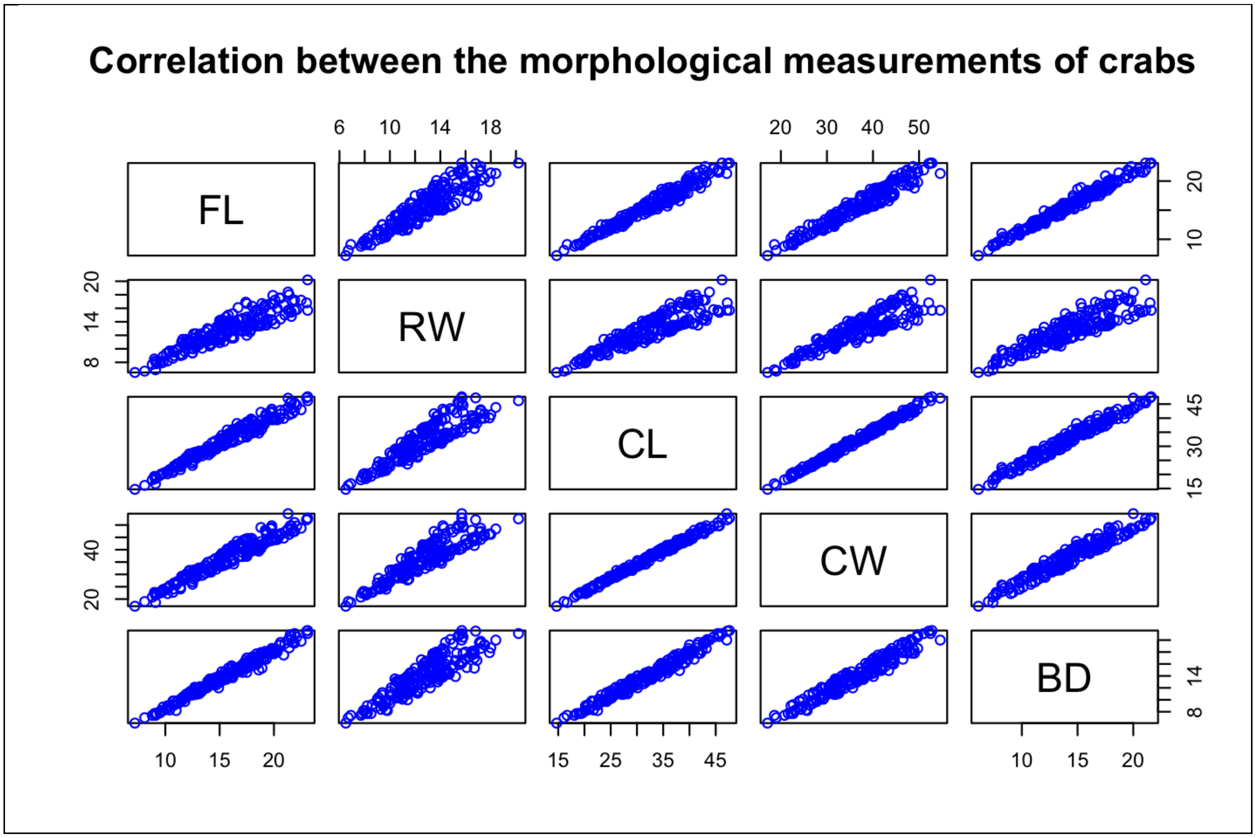
Least Sex variation: The Carapace Width show the least variation.

QUESTION FIVE (a)

Code:

```
crab_measurements<-crabs[,4:8]
pairs(crab_measurements, col= "blue", main =
        "Correlation between the morphological measurements of
crabs")
```

Output:



QUESTION FIVE (b)

<u>FL AND RW</u>

There is a strong positive relationship.

<u>FL AND CL</u>

There is a strong positive relationship.

<u>FL AND CW</u>

There is a strong positive relationship.

<u>FL AND BD</u>

There is a strong positive relationship.

RW AND CL

There is a strong positive relationship.

RW AND CW

There is a strong positive relationship.

RW AND BD

There is a strong positive relationship.

CL AND CW

There is a strong positive relationship.

CL AND BD

There is a strong positive relationship.

CW AND BD

There is a strong positive relationship.

QUESTION SIX

The report analyzes morphological measurements of 200 crabs collected in Fremantile, Australia, examining variations between species and sex.

Findings

The dataset had a balanced dataset with 100 blue crabs, 100 orange crabs, 100 female crabs and 100 male crabs.

Major Observations

- Orange crabs are consistently larger than Blue crabs across all measurements
- Greatest species variation: Carapace Length (+4.1 mm)
- Least sex variation: Carapace Width (1.2 mm difference)
- All morphological measurements show strong positive correlations ($r > 0.8$)
- Distributions are approximately normal with minimal outliers

Conclusion

Orange crabs are larger than blue crabs while sex-based differences are smaller.